Where's Waldo?

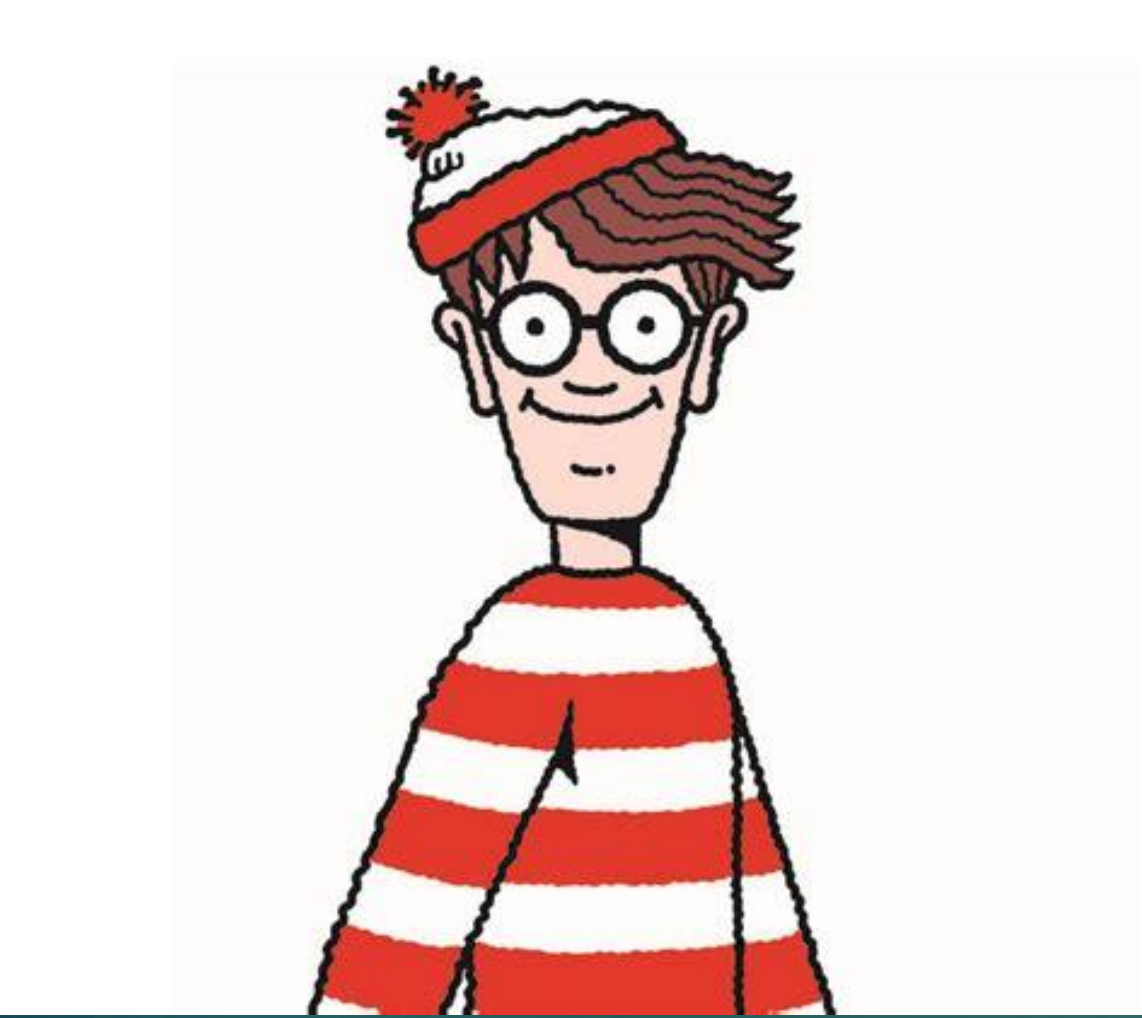
COURSE: COMP 4360

DATE: FEBRUARY 3RD, 2026

FOR: DR. CHRISTOPHER HENRY

BY: RUDY KISSEL, SATYAM BHANOT, & GABRIEL BARNABE

Problem Domain & Dataset



Source: https://www.animationmagazine.net/2011/11/classics-wheres-waldo-finds-its-way-to-mgm/ [4]

Have you seen this man?

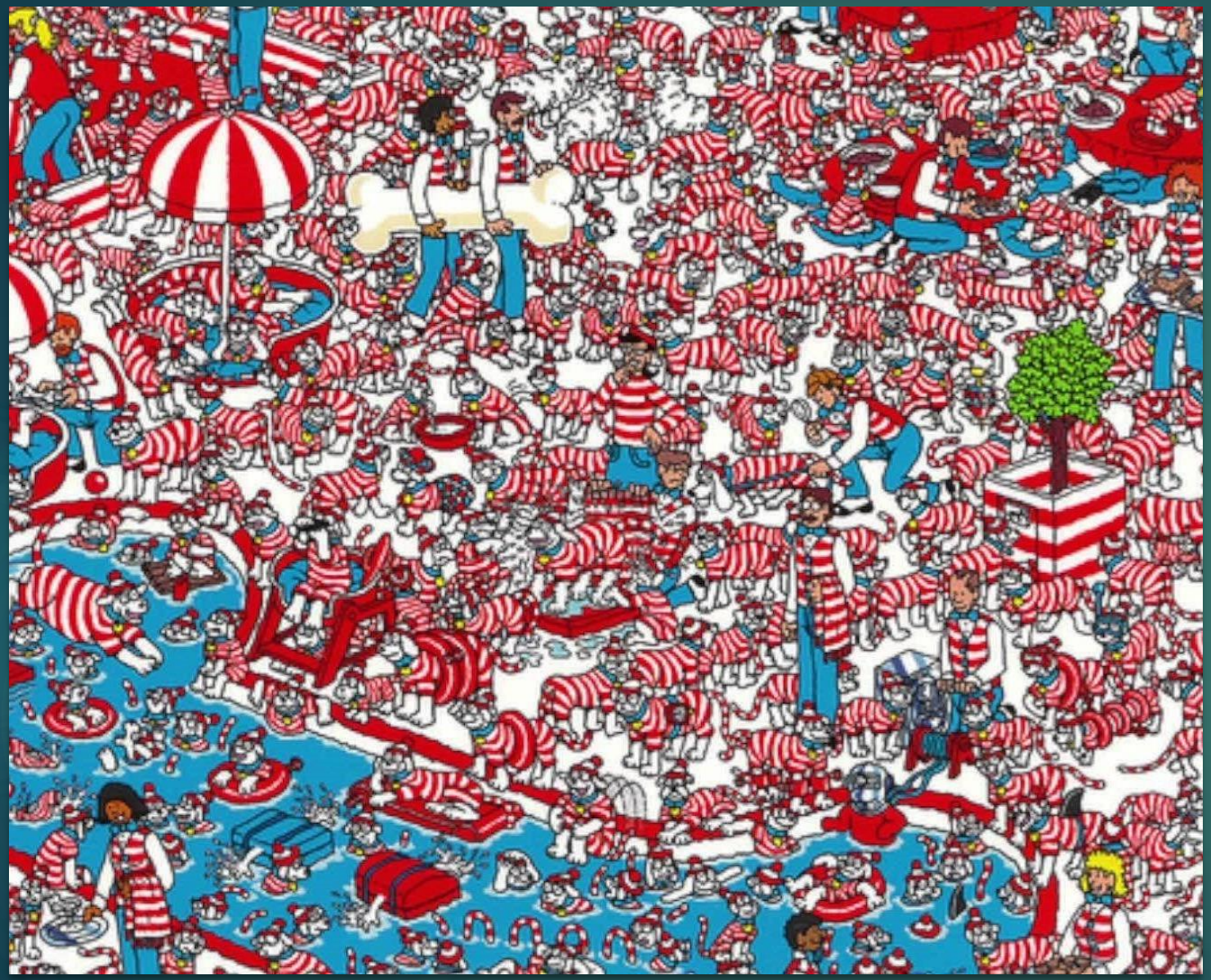
Dataset

Our Dataset

- 19 original images
- A python script was run to generate smaller labelled versions of these images
- There are color, grayscale and black and white versions of these images available
- Format is .jpg
- 317 (256 x 256) images
- 1344 (128 x 128) images
- 5376 (64 x 64) images

Original Dataset

- Manually processed video frames taken by one person in a field of strawberries
- Profoundly different



Source: https://wallpaperaccess.com/wheres-waldo [5]

Why is finding Waldo so dang hard?

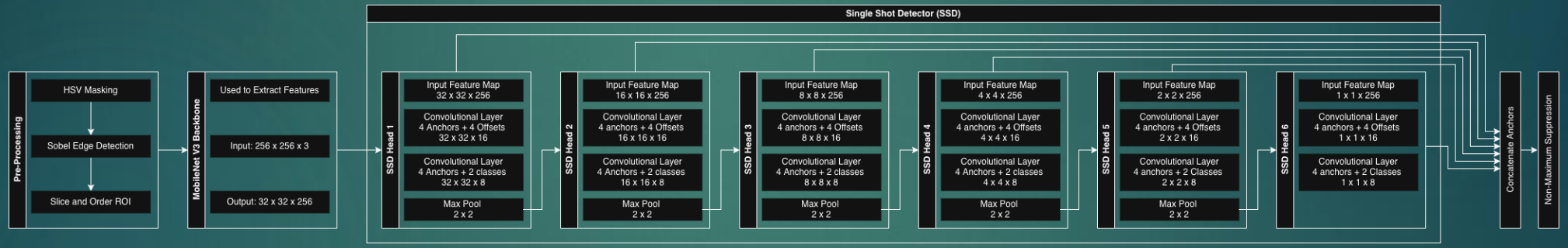
Source Paper

Title: A Strawberry Detection System Using Convolutional Neural Networks

Location: 2018 IEEE International Conference on Big Data (Big Data)

Source: See [1]

Architecture Diagram



Adaptation Strategy

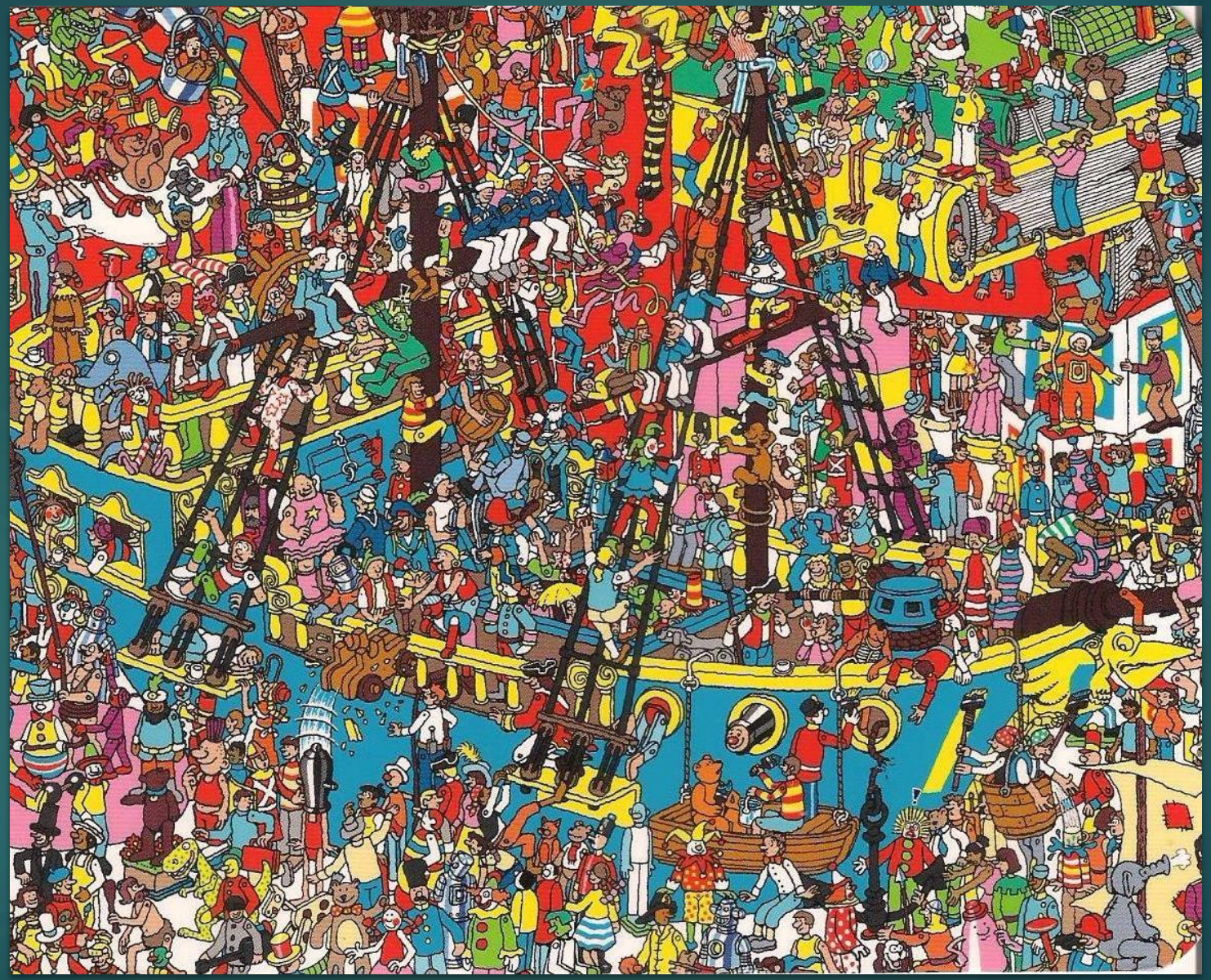
A LITTLE GAME

Find the strawberry.



Source: https://www.learnaboutnature.com/plants/fruits/strawberry/ [3]

Now, find the strawberry.



Source: Gabe, but also https://github.com/tadejmagajna/HereIsWally [2]



Source: 100% Gabe

Well, it's right here.

Adaptation Strategy

- Original paper used colour masking to create regions of interests (ROIs)
- Had few layers for feature map generation
- We introduce an alternate method for generating ROIs (Image Sub-Sectioning -> Dual HSV Masking -> Sobol Edge Detection -> Density Calculation)
- Waldo can be small, so we need more layers to extract finer features

Development Plan

Environment

Tensor Flow

Benefits

- Simple high-level API
- Faster to prototype basic models
- Large number of built-in training utilities

Drawbacks

- Harder to modify internal layers once the model is built
- More difficult to bypass or remove individual components
- Ablation studies often require rewriting parts of the model
- Less flexible when working with custom data pipelines and intermediate outputs

Why We Did Not Choose It

- Our project requires multiple ablation studies
- We need to easily disable components such as the Sobel edge detector
- TensorFlow is less convenient for fine-grained architectural changes

PyTorch

Benefits

- Easy to modify and experiment with model structure
- Individual layers and components can be turned on or off
- Simpler to inspect intermediate outputs during training
- Better suited for controlled ablation studies
- Flexible data loading for multiple image resolutions and formats

Drawbacks

- Slightly more setup code compared to Keras
- Requires more manual handling of training loops

Environment

We chose PyTorch!

- Our project focuses on **understanding model behavior**, not just training accuracy

- We need to:
  - Remove layers
  - Bypass the Sobel edge detector
  - Change input formats and resolutions

- PyTorch allows us to make these changes cleanly and safely
- The extra setup cost is acceptable given the flexibility it provides

Resources

- **Primary hardware:** NVIDIA GTX 1080 Ti

  Sufficient memory for the dataset and model size

  Allows multiple training runs for ablation studies

- **Fallback options:** Google Colab

  University GPU resources (if required)

This setup ensures hardware limitations do not prevent experimentation

Ablation Plans

Sobel Edge Detector Bypass

- Test with:
  - Sobel edge detector enabled
  - Sobel edge detector bypassed
- Why? Edges highlight outlines and repeated patterns
- Failure case: Reduced performance without edge information

Model Size Reduction

- Reduce:
  - Number of convolutional layers
  - Number of filters per layer
- Why? Detecting small objects in cluttered scenes may require higher capacity
- Failure case: Model predicts “No Waldo” when there is, in fact, a Waldo

Ablate (?) More!

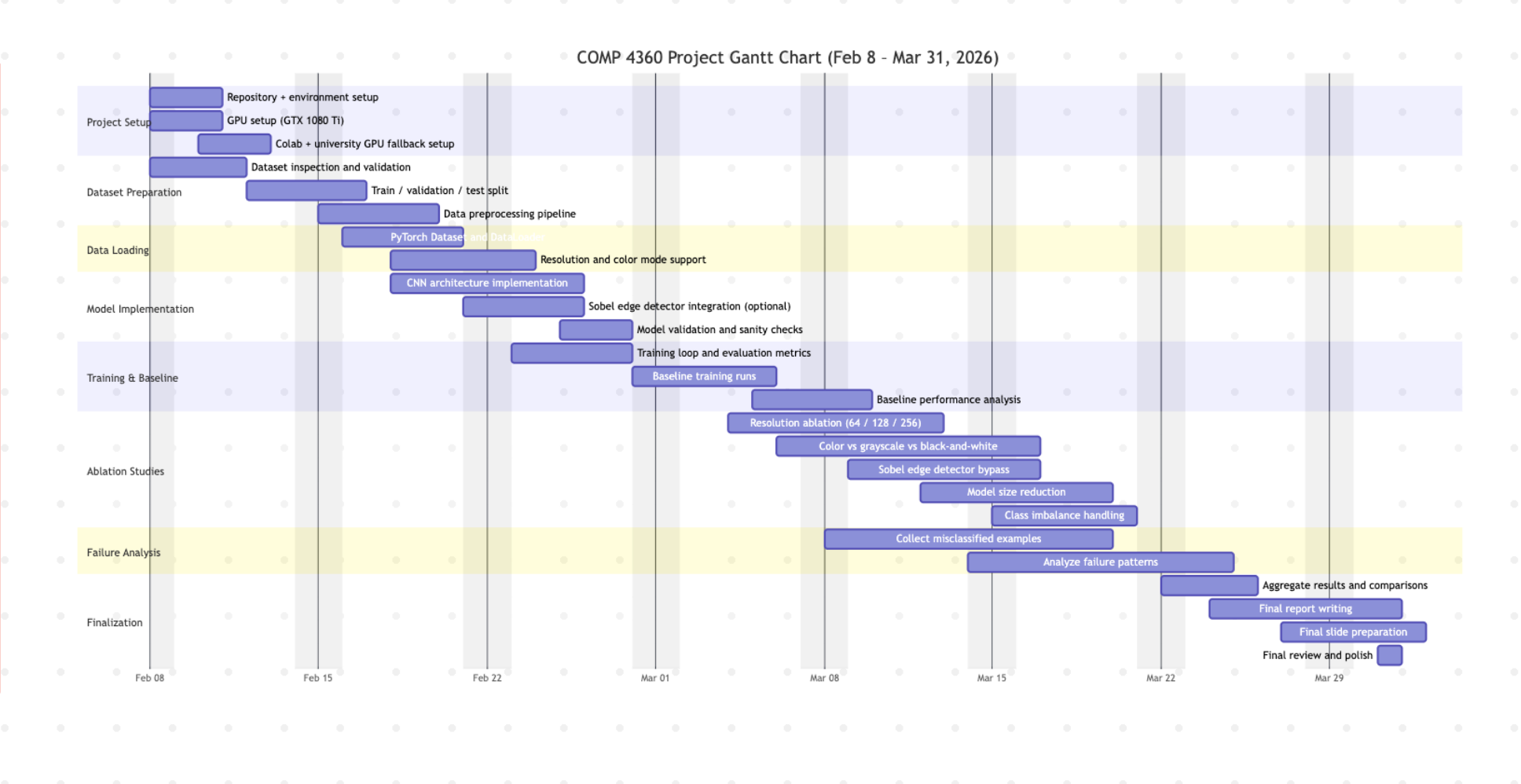
Image Resolution

- Train and evaluate the model using:
  - 64×64 images
  - 128×128 images
  - 256×256 images
- This tests:
  - Sensitivity to the size of Waldo in the image
  - Dependence on fine visual details
- Why? Waldo is often small and partially visible
- Failure case: Increased false negatives at lower resolutions

Color Information

- Compare performance using:
  - Color images
  - Grayscale images
  - Black-and-white images
- This tests:
  - Dependence on color patterns
- Why? Waldo's red and white stripes are distinctive
- Failure case: Large performance drop when color is removed

Gantt Chart



Source: 100% Satyam

Role Division

Role Division

There are 3 main components to our project:

1. Implementation of the CNN model on Pytorch - Gabe will be responsible for leading this initiative
2. Training of CNN model on Pytorch - Rudy will be responsible for leading this initiative
3. Ablation studies - Satyam will be responsible for leading this initiative

Note: This is dynamic, as we will all contribute to some extent with every component

EDI Considerations

EDI Considerations

- Believe it or not, not everyone looks like Waldo
- For typical computer vision, it's important to train data on people of as many different kinds as possible to prevent introducing biases
- Models with biases perpetuate biases

Fin.

Citations

- [1] N. Lamb and M. C. Chuah, "A Strawberry Detection System Using Convolutional Neural Networks," *2018 IEEE International Conference on Big Data (Big Data)*, Seattle, WA, USA, 2018, pp. 2515-2520, doi: 10.1109/BigData.2018.8622466.
- [2] tadejmagajna, "GitHub - tadejmagajna/HereIsWally: Deep learning project that solves Where's Wally puzzles by finding Wally in an image," *GitHub*, 2025. https://github.com/tadejmagajna/HereIsWally (accessed Feb. 02, 2026).
- [3] R. Jones, "Strawberry - Learn About Nature," *Learn About Nature*, Jul. 27, 2016. https://www.learnaboutnature.com/plants/fruits/strawberry/ (accessed Feb. 02, 2026).
- [4] Ramin Zahed, "Classic's Where's Waldo? Finds Its Way To MGM," *Animation Magazine*, Nov. 08, 2011. https://www.animationmagazine.net/2011/11/classics-wheres-waldo-finds-its-way-to-mgm/ (accessed Feb. 02, 2026).
- [5] "Awesome Where's Waldo Wallpapers - WallpaperAccess," *WallpaperAccess*, 2026. https://wallpaperaccess.com/wheres-waldo (accessed Feb. 02, 2026).